OFFLINE-AWARE · REQUEST-CENTRIC · ACID VERIFIED

EDITH

Emergency Dispatch & Intelligence Tracking Hub

"From fragmented emergency updates to accountable healthcare coordination."

CORE CAPABILITY
Offline-First Outbox
Service Worker + IndexedDB outbox preserves field requests without network.

OPERATIONAL MODEL
Request-Centric Model
Facilities volunteer spare resources on demand; zero forced inventory exposure.

DATA INTEGRITY
Atomic Row-Locking
PostgreSQL ACID transactions eliminate over-allocation in race conditions.

NON-REPUDIATION
Two-Phase Verification
Driver drop-off does not equal delivery; recipient physician must verify receipt.

TEAM JARVIS
Rathinam Technical Campus · Coimbatore

PROJECT ROSTER
Purusouthanan B (Lead) · Sakthivel R · Sanjo Thomas · Sheik Arsh S

CORE TECHNOLOGY STACK
FastAPI · PostgreSQL 16 · Vanilla PWA

Meet Team JARVIS · Rathinam Technical Campus

Multidisciplinary student engineering team dedicated to solving disaster-zone healthcare coordination breakdown.

TEAM LEAD · RTC-CSE

Purusouthanan B

Product & Systems Architect

Architects the end-to-end platform architecture, disaster request state machines, role-based security boundaries, and offline sync contracts.

`Focus: State Machine Integrity & System Architecture`

ENGINEERING · RTC-CSE

Sakthivel R

Backend & API Engineer

Engineers FastAPI modular monolith services, Pydantic schema validation boundaries, RESTful API endpoints, and server-side error recovery.

`Focus: Modular Monolith Services & API Reliability`

CLIENT SYSTEMS · RTC-CSE

Sanjo Thomas

Frontend & Offline Systems

Develops the lightweight PWA application shell, Service Worker cache strategies, IndexedDB local outbox, and automatic sync reconnection manager.

`Focus: PWA Caching & Client-Side Outbox Sync`

INTEGRITY & QA · RTC-CSE

Sheik Arsh S

Data Integrity & Verification

Manages PostgreSQL relational constraints, row-level concurrency locks, idempotency verification tables, and disaster simulation chaos testing.

`Focus: Concurrency Locking & Verification Suites`

TEAM JARVIS · CORE ENGINEERING PHILOSOPHY

1. Ground-Truth Analysis: Studied historical failure modes in Tamil Nadu cyclone and flood logistics breakdown.

2. Deliberate Simplicity: Avoided fragile microservice bloat in favor of a robust, highly maintainable modular monolith.

3. Zero False Guarantees: Never present un-synced offline drafts as confirmed server records.

4. Measurable Accountability: Enforce strict non-repudiation: every status shift has an immutable human actor record.

Why This Problem Matters: Healthcare in Catastrophic Disasters

In severe floods and cyclones, the bottleneck is rarely total supply absence, but a failure of coordination.

1. Demand Surges & Depletion

ACUTE RISK

Floods and cyclones cause sudden, localized spikes in trauma, drowning, electrocution, waterborne contagions, and hypothermia across relief centres.

Supplies Depleted in Hours: Medical oxygen cylinders, O-negative blood units, tetanus toxoid, pediatric IV saline, and generator diesel exhaust rapidly with zero visibility into regional backups.

2. Fragmented Channel Chaos

COORDINATION GAP

Hospitals and medical camps broadcast desperate appeals across disjointed WhatsApp groups, SMS threads, informal phone calls, and paper slips.

Fatal Distribution Asymmetry: A high-profile urban shelter receives redundant donations from 4 different donors, while an isolated medical camp 5 km away receives nothing.

3. Severed Infrastructure

NETWORK DROP

High cyclone winds topple mobile towers; flooded electrical substations plunge districts into darkness; fiber backhauls are severed by uprooted trees.

Cloud SaaS Fragility: Standard enterprise hospital management systems, EHRs, and heavy SaaS dashboards freeze or crash the moment 4G/5G drops.

ANATOMY OF DISASTER COORDINATION COLLAPSE (CURRENT REALITY)

STAGE 1	STAGE 2	STAGE 3	STAGE 4	OUTCOME
Surge Demand	**Blind Shouting**	**No Assigned Owner**	**Network Drop**	**Unverified Zero**
Camp runs out of critical pediatric IV saline	Needs blasted into 6 unmonitored WhatsApp groups	Everyone assumes someone else is arranging supplies	Tower battery fails; camp loses cellular connectivity	Zero accountability; supplies never arrive at destination

The Problem Statement: Coordination Without Accountability

Formalizing the critical gap in emergency healthcare response during severe weather disasters.

FORMAL HACKATHON PROBLEM STATEMENT

"During severe floods and cyclones, hospitals and medical camps struggle to coordinate emergency resource requests because communication channels are fragmented, availability data becomes stale, ownership is unclear, and network disruptions interrupt cloud-dependent workflows."

PAIN POINT 01

Fragmented Comm Channels

Calls, WhatsApp, SMS, and paper memos create isolated silos with zero unified operational picture.

PAIN POINT 02

Stale Resource Availability

Static spreadsheets and verbal promises don't reflect rapid field depletion or prior commitments.

PAIN POINT 03

Unassigned Ownership

Urgent requests broadcast into groups sit unacknowledged without an assigned human coordinator.

PAIN POINT 04

Unverified Fulfilment

"Dispatched" is conflated with "Delivered". Lost, diverted, or damaged cargo goes completely undetected.

PAIN POINT 05

Cloud-Only Fragility

Standard web applications freeze or throw connection errors the moment infrastructure loses power.

Target Scope Clarification: EDITH does not replace existing hospital ERPs or medical triage. It targets the acute coordination vacuum between independent facilities during sudden regional crises.

DISASTER COORDINATION FOCUS

Existing Coordination Gap vs. The EDITH Standard

Moving from informal, fragile communication to structured, resilient accountability.

Coordination Dimension	Informal Channels (Phone / WhatsApp)	Generic Dashboards / Spreadsheets	EDITH Platform Standard
Request Visibility	Siloed in private chats; no unified operational view	Passive tabular lists; zero real-time urgency sorting	**Unified regional triage queue with dynamic priority levels**
Ownership & SLA	No assigned coordinator; broadcast without accountability	Ambiguous ticket states; unmonitored ticket queues	**Mandatory ICC Coordinator assignment with age timers**
Resource Offers	Verbal promises in chats; unverified volunteer claims	Static inventory lists disconnected from active requests	**Structured voluntary offers tied explicitly to active needs**
Allocation Tracking	Severe double-promising risk across separate callers	Manual cell edits; vulnerable to simultaneous overwrites	**ACID row-level locks preventing over-allocation race conditions**
Partial Fulfilment	Lost in conversation history; requires manual re-asks	Binary 'Complete / Incomplete' state flags	**Automated state splitting: fulfilled quantity vs remaining need**
Delivery Confirmation	Assumed delivered upon driver dispatch call	One-sided driver click with zero recipient sign-off	**Two-phase verification: Recipient physician confirms count**
Offline Operation	Fails completely when tower battery dies or signal drops	Browser error screens; lost form drafts on submit	**Service Worker cache + IndexedDB outbox queue + retry engine**
Auditability & History	Ephemeral messages easily deleted or buried	Basic updated_at timestamp without actor identity	**Immutable append-only audit trail tracking actor, time, & diff**

Core Architectural Takeaway: EDITH replaces informal guesswork with guaranteed operational state machines. Every resource moving across the disaster zone has an authenticated origin, an assigned coordinator, and a verified destination.

FORMAL GUARANTEES

The EDITH Operational Lifecycle: 7-Stage Accountable Coordination

Every emergency request moves through an explicit, state-machine-governed progression with non-repudiation.

STAGE 1
SUBMITTED
Medical camp drafts urgent need (e.g. 20 O2 cylinders)
CAMP USER

STAGE 2
ACKNOWLEDGED
ICC coordinator reviews & takes operational ownership
ICC COORD

STAGE 3
OFFER MADE
Nearby hospital volunteers available surplus (12 units)
DONOR FACIL

STAGE 4
ASSIGNED
ICC authorizes allocation; DB executes row-level lock
ATOMIC LOCK

STAGE 5
IN TRANSIT
Logistics packages cargo, logs vehicle ID & dispatches
LOGISTICS

STAGE 6
DROP-OFF
Driver logs physical arrival at destination gates
DROP-OFF ONLY

STAGE 7
CONFIRMED
Recipient physician verifies received count & condition
CLOSED/RESOLVED

EXCEPTION PATH A
Partial Fulfilment Split
If 12 of 20 units are verified, 12 units resolve while the remaining 8 units automatically stay active in the regional triage queue for secondary donor offers.

EXCEPTION PATH B
Delivery Dispute Protocol
If cargo arrives damaged or missing, the recipient flags a formal dispute. EDITH halts resolution, alerts ICC coordinators, and creates an audit investigation ticket.

EXCEPTION PATH C
Reassignment & Pivots
If floodwaters submerge transit roads, coordinators can reassign cargo to alternate staging hubs with complete historical state logging.

Offline Resilience: The EDITH Difference

Preserving operational continuity during network blackouts — without corrupting central state.

STEP 1
Online Baseline
Normal HTTPS connection to FastAPI backend.
CONNECTED

STEP 2
Connection Lost
Tower battery dies; PWA service worker intercepts calls.
OFFLINE

STEP 3
Local Capture
Doctor drafts request; saved instantly to IndexedDB outbox.
INDEXEDDB

STEP 4
Explicit UI Truth
UI displays amber tag: "Queued Locally · Pending Sync".
PENDING SYNC

STEP 5
Signal Returns
Sync manager flushes outbox with Idempotency Key.
REPLAY QUEUE

STEP 6
Server Commit
FastAPI validates RBAC, writes DB, returns canonical ID.
CANONICAL ID

No False Mesh Claims
REALISTIC ARCH

Separate disconnected devices cannot magically sync across floodwaters without an active radio or physical communication path. EDITH never makes false claims of peer-to-peer miracles.

Truth Transparency
UI INTEGRITY

A locally-saved draft is never presented to the user as a server-confirmed record. Un-synced requests carry a distinct visual warning and temporary UUID to avoid false confidence.

Idempotency Protection
ZERO DUPLICATES

When an intermittent 2G connection flickers, repeated sync retries cannot create duplicate emergency tickets or allocate supplies twice. Server-side idempotency keys enforce safe replay.

Three-Tier Modular Architecture for Resilient Coordination

Strict separation of concerns enforcing human authority, offline survival, and ACID truth.

TIER 1 · CLIENT PWA

PRESENTATION

`Vanilla HTML5 · CSS3 · Modern ES6+ JS`

- **Service Worker:** Pre-caches application shell, icons, and offline fallback screens.
- **IndexedDB Outbox:** Stores pending mutations and cached entity reads.
- **Sync Engine:** Background retry manager with exponential backoff and network probes.

```
Performance: Sub-500ms initial load on 2G connections
```

TIER 2 · API SERVICES

BUSINESS LOGIC

`FastAPI 0.115+ · Pydantic v2 · Python 3.12`

- **Modular Monolith:** Auth, Facilities, Requests, Offers, Allocations, Logistics, Sync.
- **Boundary Validation:** Strict Pydantic contracts reject malformed offline payloads.
- **Idempotency Guard:** Middleware detects duplicated network requests via UUID keys.

```
Throughput: Asynchronous non-blocking event loop
```

TIER 3 · DATA STORE

AUTHORITATIVE TRUTH

`PostgreSQL 16 (Dockerized) · SQLAlchemy 2.0`

- **ACID Guarantees:** Authoritative shared state; strict relational schema constraints.
- **Row-Level Locks:** `SELECT ... FOR UPDATE` eliminates race conditions on allocations.
- **Immutable Audit Log:** Append-only table logging actor, timestamp, and diff for every shift.

```
Deployment: Single-container reproducible image
```

AUXILIARY INTEGRATION: OPTIONAL ASSISTIVE AI SCOPE

External LLM APIs strictly limited to secondary tasks: audio transcription, structured request extraction, and triage summaries. **Zero autonomous decision-making power; fail-safe graceful degradation if offline.**

FAIL-SAFE ASSISTIVE ONLY

Deliberate Engineering Choices: Built for Field Feasibility

Every technology selected for minimal resource footprint, high resilience, and rapid deployment.

Architectural Layer	Selected Technology	Operational Justification & Field Fit
Frontend Interface	Vanilla HTML5, CSS3, Modern JS	Zero framework bloat or compilation overhead; guarantees instant load and buttery responsiveness even on low-end budget smartphones.
Offline Shell	Service Worker & Cache API	Enables offline launching, home-screen installation without app store friction, and robust fallback routing when network drops.
Client Persistence	IndexedDB (Native Browser DB)	Provides high-capacity, transactional, structured local storage for pending outbox mutations and local read caches.
Backend Framework	Python FastAPI (v0.115+)	Lightweight ASGI micro-framework with native async request pipeline, automatic OpenAPI documentation, and minimal memory usage.
Contract Validation	Pydantic v2 (Rust Core)	High-speed data boundary validation ensuring corrupted or incomplete offline payloads are caught and rejected cleanly.
Data Access / ORM	SQLAlchemy 2.0 (Psycopg 3)	Type-safe database abstraction with granular transaction control, explicit connection pooling, and row-level locking support.
Authoritative Store	PostgreSQL 16 (Dockerized)	World-class ACID durability, proven concurrency control, JSONB support for flexible audit trails, and containerized portability.
Testing & Verification	Pytest, Asyncio, Playwright	Automated test suites covering concurrent allocation race conditions, idempotency replays, and simulated offline recovery.

Stack Rationale: By relying on standard web APIs (Service Worker, IndexedDB) and a lightweight FastAPI/PostgreSQL backend, EDITH requires zero expensive licenses and runs smoothly on low-cost disaster command infrastructure.

ZERO PROPRIETARY LOCK-IN

Data Integrity, Security & Concurrency Controls

Engineering safeguards that eliminate race conditions, unauthorized actions, and data corruption.

1. Role-Based Access Control (RBAC)

AUTHORITY BOUNDARIES

Permissions are enforced strictly by backend API middleware and database foreign key checks, never merely hidden in the UI:

- **Hospital / Camp Users:** Scoped strictly to drafting requests and voluntary offers for their own registered facility.
- **ICC Coordinators:** Global zone authority to acknowledge, triage, approve allocations, and close tickets.
- **Logistics Drivers:** Scoped strictly to updating shipment progress and logging cargo drop-off.

2. Atomic Allocation & Race Prevention

ACID ROW LOCKS

Disaster Scenario: Two coordinators simultaneously attempt to allocate the final 10 oxygen cylinders from Hospital Beta to two different camps.

```
SELECT quantity_available FROM offers WHERE id = :id FOR UPDATE;
-- Row is locked; Coordinator A commits 10; remaining becomes 0.
-- Coordinator B's lock releases; detects 0 available; gracefully rejected with 409 Conflict.
```

Guarantees zero physical over-allocation under peak crisis concurrency.

3. Idempotency & Duplicate Prevention

SAFE REPLAY

When unstable cellular links drop mid-transmission, automated client retries can accidentally spawn duplicate tickets:

- Every mutation payload generates a client-side UUID v4 `X-Idempotency-Key`.
- The backend verifies this key against the `processed_operations` table before execution.
- Repeated requests return the cached original response without creating duplicate records.

4. Two-Phase Fulfilment Verification

NON-REPUDIATION

Dispatching relief cargo is not proof of arrival:

- Driver logs physical drop-off with vehicle ID and timestamp at the destination gate.
- **Resolution is NOT triggered:** Ticket remains in `DELIVERED_PENDING_VERIFICATION`.
- The receiving medical officer must log in, inspect cargo, and verify physical count.

Command-Centre Dashboard: Real-Time Operational Triage

High-density, low-cognitive-load interface designed for disaster coordinators.

INCIDENT COMMAND CENTRE · SECTOR 4 DISASTER DESK · NETWORK: CONNECTED · LAST SYNC: 3S AGO · SIMULATED TEST DATA · CYCLONE SCENARIO

TOTAL OPEN REQUESTS	UNACKNOWLEDGED NEEDS	OFFERS PENDING SIGN-OFF	IN TRANSIT / ACTIVE FLEET
34	3	6	12
8 Critical · 14 High · 12 Normal	Requires Immediate Coordinator Triage	Donor Matches Awaiting Allocation Lock	Logistics Vehicles En Route to Camps

ACTIVE EMERGENCY QUEUE (SECTOR 4)

ID	Facility	Required Items	Status	Age
REQ-0842	**Camp Alpha (Govt HSS)**	25 O2 Cylinders CRITICAL	ACKNOWLEDGED	14m
REQ-0839	Apollo Field Clinic	40 Units O- Blood CRITICAL	IN TRANSIT (15/40)	42m
REQ-0835	Camp Delta (Maternity)	500L Saline IV HIGH	OFFER RECEIVED	1h 10m
REQ-0830	PHC Kadambur	1× 15kVA Generator CRITICAL	VERIFIED RESOLVED	2h 05m

DETAILS: REQ-0842

OWNER: ICC-COORD-02

Camp Alpha (Govt HSS) · Velachery Sector
Need: 25 D-Type Oxygen Cylinders (Pediatric Emergency)

LINKED VOLUNTEER OFFER
Coimbatore Medical College Hospital
15 Oxygen Cylinders available for immediate dispatch

APPROVE ALLOCATION · PARTIAL SPLIT: 15/25

AUDIT TIMELINE: 14:02 Created · 14:08 Acknowledged by Anand M. · 14:15 Offer Logged

Demo Scenario: Oxygen Supply Recovery in Cyclone Mandous

Step-by-step resolution of a critical resource shortage amidst intermittent connectivity.

PHASE 1 · 14:02

Outage at Camp Beta

Cyclone knocks out power & cellular links at Camp Beta. Dr. Priya identifies urgent need for 20 oxygen cylinders.

INDEXEDDB DRAFT

PHASE 2 · 14:18

Transient Signal Sync

A satellite telecom van passes nearby, restoring 2G for 40 seconds. EDITH sync manager flushes with Idempotency Key `idemp_91a8`.

COMMITTED REQ-0512

PHASE 3 · 14:25

ICC Triage & Offer

ICC Commander acknowledges REQ-0512. City Hospital logs voluntary offer: 12 cylinders available for immediate dispatch.

VOLUNTARY OFFER

PHASE 4 · 14:34

Atomic Lock & Transit

ICC approves allocation. DB row-lock locks 12 units (0 over-allocation). Logistics vehicle TN-01-G-7712 departs with driver.

IN TRANSIT

PHASE 5 · 15:45

Verified Receipt

Driver drops cargo. Dr. Priya inspects and verifies 12 units. State splits: 12 confirmed closed, 8 units remain active in queue.

VERIFIED SPLIT

What This Drill Proves: No data was lost during blackout; zero duplicate tickets created on transient reconnection; zero over-allocation occurred; and unfulfilled quota (8 cylinders) was not forgotten.

SIMULATED DRILL VALIDATION

Core Capabilities & Pilot Validation Metrics

Translating architectural resilience into measurable life-saving coordination outcomes.

1. Operational Visibility

- Unified multi-facility triage queue with dynamic urgency badges.
- Mandatory coordinator assignment eliminating orphaned requests.
- Automated SLA timers monitoring unacknowledged critical emergencies.
- Immutable audit history documenting every transition.

2. Resource Coordination

- Sovereign voluntary offers without exposing total facility inventory.
- Transactional atomic allocation eliminating double-promising.
- Automatic state-splitting for partial fulfilments.
- Transit tracking and formal delivery dispute escalation.

3. Connectivity Resilience

- Offline application shell caching via PWA Service Worker.
- IndexedDB transactional outbox buffer preserving field drafts.
- Idempotent sync engine preventing duplicate tickets on retry.
- Two-phase recipient verification separating dispatch from delivery.

TARGET PILOT VALIDATION HYPOTHESES (MEASURABLE SUCCESS CRITERIA)

TIME-TO-ACK
>65% Faster
Reduction in unacknowledged request duration

ACCOUNTABILITY
100% Owned
Every emergency linked to a named coordinator

OVER-ALLOCATION
0 Errors
Zero double-commit errors under concurrent load

OFFLINE REPLAY
100% Synced
Zero data loss across simulated network drops

VERIFICATION
100% Verified
Zero tickets resolved without recipient sign-off

Institutional Adoption Model & Sustainability Pathway

A viable deployment model for humanitarian agencies and public health authorities.

Target Customer Ecosystem

- **State Disaster Management Authorities (SDMAs):** Central operators of Incident Command Centres (ICCs).
- **Public & Private Hospital Networks:** Secondary supply hubs offering voluntary surplus resources.
- **Relief NGOs (Red Cross, MSF, Trusts):** Field operators managing temporary relief clinics.
- **District Emergency Operation Centres (DEOCs):** Regional coordinators managing logistics fleets.

Sustainability & Operations

- **Open-Core Public Good:** Core coordination software is auditable to foster inter-agency trust.
- **Annual Institutional SLA & Hosting:** Managed cloud/edge deployments, automated failover, and disaster-standby support.
- **Simulation Drills & Onboarding:** Pre-disaster simulation workshops, tabletop exercises, and staff training.
- **Zero Frontline Barrier:** 100% free for temporary camps and frontline volunteer doctors.

Pragmatic Adoption Pathway

- **Phase 1 · Academic & Hackathon Pilot:** Simulated multi-facility disaster drills in Coimbatore district.
- **Phase 2 · Tabletop NGO Simulation:** Partner with local humanitarian trusts to test camp workflows.
- **Phase 3 · District Mock-Drill Integration:** Shadow municipal flood preparedness exercises.
- **Phase 4 · Institutional Deployment:** Formal SDMA accreditation for active monsoon seasons.

Mitigating Public Sector Friction: Avoiding long procurement cycles by providing a plug-and-play Docker image and zero-login guest drafting for emergency field responders during acute crisis activations.

ZERO-FRICTION DEPLOYMENT

Technical Feasibility, Testing Matrix & Risk Safeguards

Rigorous software engineering that separates practical innovation from speculative concepts.

Comprehensive Testing Matrix

- **Synthetic Disaster Load Testing:** Scripted simulation of 100+ concurrent requests across 20 simulated medical relief camps.
- **Concurrency Race Condition Tests:** Automated Pytest suite firing 10 simultaneous allocations against a single remaining stock pool.
- **Network Chaos Verification:** Automated simulated connection dropouts during active outbox flushes to ensure idempotent recovery.
- **State Transition Coverage:** 100% automated branch testing across all 7 lifecycle states and exception paths.

Operational Risk vs. Mitigation

Risk: Stale Resource Information
Mitigation: Request-centric model; facilities volunteer only verified unallocated surplus.

Risk: Multi-Device Conflict & Overwrites
Mitigation: Optimistic locking with version integers; server rejects stale updates.

Risk: High-Stress Field Adoption Resistance
Mitigation: Ultra-lightweight form design requiring under 30 seconds to submit.

HARDWARE REQUIREMENT
Minimal Infrastructure
Runs smoothly on $20/mo cloud VM or local edge laptop

ZERO LICENSING FEES
100% Open Standards
Zero proprietary vendor lock-in or recurring API costs

FIELD COMPATIBILITY
Universal Browser Support
Operates on Android, iOS, Windows, Linux, and macOS

Execution Roadmap & The EDITH Commitment

From hackathon baseline to hardened institutional disaster deployment.

PHASE 1
Core Coordination
Authentication, RBAC boundaries, request lifecycle state machine, and append-only audit trail.

BASELINE READY

PHASE 2
Resource Flow
Voluntary offers, atomic row-lock allocation, logistics drop-off, and verified fulfilment.

BASELINE READY

PHASE 3
Offline Resilience
Service Worker app-shell cache, IndexedDB outbox queue, and idempotent background sync.

BASELINE READY

PHASE 4
Hardening & QA
Network chaos testing, high-concurrency race stress tests, and tabletop disaster drills.

PLANNED Q1

PHASE 5
Assistive AI
Audio speech-to-text intake, automated request summarization, and duplicate ticket suggestions.

FUTURE SCOPE

"EDITH turns fragmented emergency requests into accountable, traceable coordination — designed for the moments connectivity cannot be taken for granted."

TEAM JARVIS · RATHINAM TECHNICAL CAMPUS · PURUSOUTHANAN B · SAKTHIVEL R · SANJO THOMAS · SHEIK ARSH S